INCH-POUND

MIL-PRF-1/852E
17 October 2008
SUPERSEDING
MIL-E-1/852D
23 March 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 5933WA

This specification is approved for use by all Departments and Agencies of the Department of Defense

Inactive for new design after 30 April 1997.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF -1.

DESCRIPTION: Pentode, beam power amplifier, F1 = 50 MHz, F2 = 125 MHz

Outline
Base — See figure 1
Cathode --- Coated unipotential
Base connections:

Pin No.	---	1	2	3	4	5	Cap
Element	---	h	g2	g1	k,g3	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pp
Unit:	V	V dc	V dc	V dc	mA dc	mA dc	W
Maximum:							
Class B, AF:	6.6	600	---	300	120	---	25
Class B, RF:	6.6	600	---	300	80	---	25
Class C, Telep:	6.6	475	-200	300	83	5	16.5
Class C, Teleg:	6.6	600	-200	300	100	5	25
Minimum:	6.0	---	---	---	---	---	---
TEST CONDITIONS:	6.3	600	-29	300	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Pg2	Pi	Ehk	TE	Modulation	Alt
Unit:	W	W	V	°C	---	ft
Maximum:						
Class B, AF:	3.5	60	135	160	---	60,000
Class B, RF:	2.5	37.5	135	160	---	60,000
Class C, Telep:	2.5	40	135	160	Anode	60,000
Class C, Teleg:	3.5	60	135	160	---	60,000
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification – Not Required.

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TABLE I. Testing and conditions.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance Level 8/	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 1							
1301	Heater current			0.65	If	810	990	mA
1336	Heater-cathode leakage			0.65	Ihk	---	100	µAdc
1266	Total grid current	1		0.65	Ic1	0	-4.0	µAdc
1256	Electrode current (1) (anode)	1		0.65	Ib	24	48	µAdc
1256	Electrode current (2) (anode)		Ec1 = -100 Vdc	0.65	Ib	---	500	µAdc
1236	Power oscillation		Ec2 = 200 V dc; Rg1 = 0.01 MegΩ; Ib = 100 mAdc (max); Ic1 = +mAdc; F = 15 MHz; Eccl = 0	0.65	Po	33	---	w
1201	Short and discontinuity detection			0.4	---	---	---	---
	Conformance inspection, part 2							
1211	Insulation of electrodes		Ep-all = -500 Vdc; Eg1-all = -300 Vdc	2.5	R	50	---	MegΩ
1256	Electrode current (screen)			2.5	Ic2	0	3.0	mAdc
1266	Grid emission	2	Ef = 7.5 V; Ec1 = -70 V dc	2.5	Ic1	0	-6.0	µAdc
1306	Transconductance (1)		Eb = Ec2 = 250 V dc; Ec1 = -15 Vdc	6.5	Sm	4,500	6,500	µmhos
1306	Transconductance (2)		Ef = 5.7 V; Eb = Ec2 = 250 Vdc; Ec1 = -15 Vdc	2.5	$\frac{\Delta Sm}{Ef}$	---	10	%
1266	Primary grid emission (screen)	4	Eb = Ec2 = 0; Eg2 = 175 Vac	6.5	Isc2	0	-750	µAdc
1331	Direct-interelectrode capacitance		Shield No. 312 No shield No shield	6.5	Cg1p Cin Cout	--- 8.0 5.3	0.2 14 8.7	pF pF pF
1002	Barometric pressure, reduced		Pressure = 55±5 mm Hg; Voltage = 700 Vac	6.5	---	---	---	---
1031	Low-frequency vibration		Eb = 250 Vdc; Ec1 = -10 Vdc; Ec2 = 100 Vdc; Rp = 2,000 ohms	2.5	Ep	---	250	mVac

See notes at end of table I.

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TABLE I. Testing and conditions - Continued.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance Level 8/	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 2 - Continued							
1041	Shock	6	450G; Ehk = +135 V dc	6.5	---	---	---	---
1031	Vibration fatigue			6.5	---	---	---	---
---	Shock and vibration fatigue test end points:							
1031	Low-frequency vibration			---	Ep	---	500	mA dc
1336	Heater-cathode leakage			---	Ihk	---	100	µA dc
1236	Power oscillation (change in individual tubes)			---	ΔPot	---	20	%
1111	Base pin solder depth			---	---	---	---	---
1101	Secureness of base, cap, or insert			---	---	---	---	---
1216	Base material insulating quality			---	---	---	---	---
1105	Permanence of marking			---	---	---	---	---

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = +135 V dc Ec1 = Ec2 = Eb = 0; 1 min on, 4 min off	---	---	---	---
1516	Stability life		Ehk = 135 V dc; Rg1 = 0.01 Meg Ω; TA = room	---	---	---	---
---	Stability life-test end point (1 hour):						
1236	Power oscillation (change in individual tubes)			ΔPot	---	20	%
1501	Intermittent Life	3, 7	Ehk = 135 V dc; Rg1 = 0.01 Meg Ω; TA = room TE = 160°C (min)	---	---	---	---

See notes at end of table I.

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TABLE I. Testing and conditions - Continued.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3 - Continued.						
	Intermittent life-test end points (500 hours):						
1266	Total grid current			Ic1	0	-6.0	µAdc
1301	Heater current			If	800	1,010	mA
1236	Power oscillation (Ef = 5.7 V)			ΔPo Ef	---	15	%
1236	Power oscillation (change in individual tubes)			ΔPo t	---	20	%
1236	Power oscillation (overage change)			Avg ΔPo t	---	15	%
1336	Heater-cathode leakage			Ihk	---	150	µAdc
1211	Insulation of electrodes			R	25	---	MegΩ
---	Total defectives			---	---	---	---
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic1	0	-6.0	µAcc
1301	Heater current			If	800	1,010	mA
1236	Power oscillation (change in individual tubes)			ΔPo t	---	25	%
1236	Power oscillation (average change)			Avg ,ΔPo t	---	20	%
1336	Heater-cathode leakage			Ihk	---	150	µAdc
1211	Insulation of electrodes			R	25	---	MegΩ
---	Total defectives			---	---	---	---

NOTES:

1. This test shall be performed at the conclusion of the holding period.

2. Prior to this test, the tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Eb	Ec1	Ec2	Rg1
V	V dc	V dc	V dc	Meg Ω
7.5	600	-29	300	0.01

3. Envelope-temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

4. The test duration shall be of sufficient length to obtain a stabilized negative Isc2 value. Adjust Ec1 (0 to 6 V dc) to give Pg2 = 5 W. Pg2 shall be calculated as 2.48 times the product of the rectified current and rectified voltage. A protective resistor of 15,000 ohms shall be placed in series with the primary emission current meter.

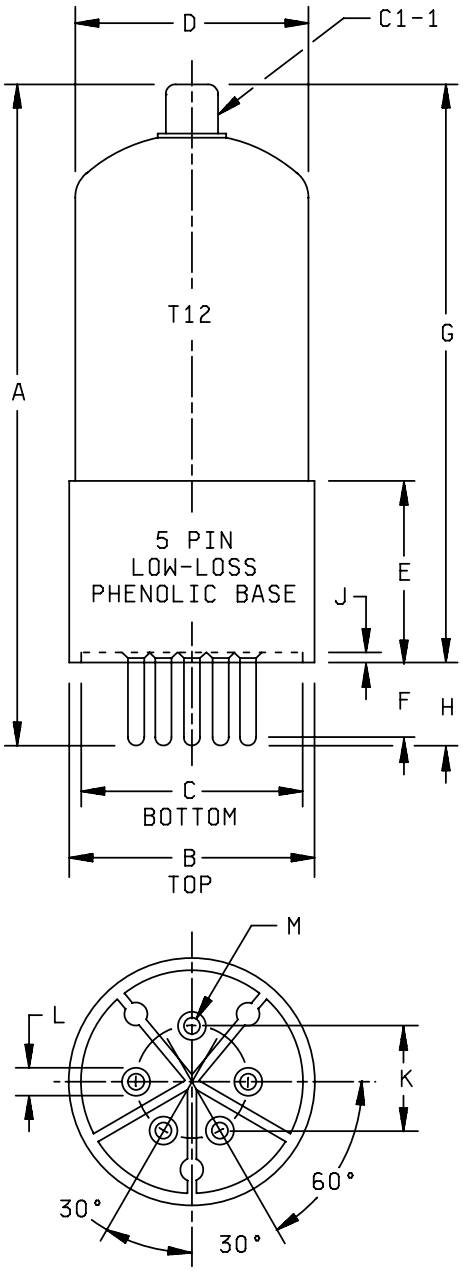
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. ANSI/ASQ Z1.4, sample size code letter E, shall apply.

6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.

7. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

8. This specification sheet utilizes accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

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Ltr	Inches		mm	
	Min	Max	Min	Max
Conformance inspection, part 1				
A	---	4.688	---	119.08
B	1.655	1.700	42.04	43.18
C	1.630	1.655	41.40	42.04
D	---	1.562	---	39.67
E	1.245	1.265	31.62	32.13
F	.430	---	10.92	---
G	3.750	4.062	95.25	103.17
H	---	.562	---	14.27
J	.075	---	1.91	---
K	.750 Nom		19.05 Nom	
L	---	.195	---	4.95
M	.122	.128	3.10	3.25

FIGURE 1. Outline drawing of electron tube type 5933 WA.

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Referenced documents. In addition to MIL-PRF-1, this document references the following: MIL-STD-1311 and ANSI/ASQ Z1.4.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project No. 5960-2008-068)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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